TIC 300146496

Eclipsing binary candidate dossier - Star Vetter

CLEAN

```
BLS EPHEMERIS
  Period       2.011353 d
  Epoch (t0)   1438.3349 BTJD
  Duration     0.0580 d
  Depth        2.231 %  (22,312 ppm)
  SNR          42.4

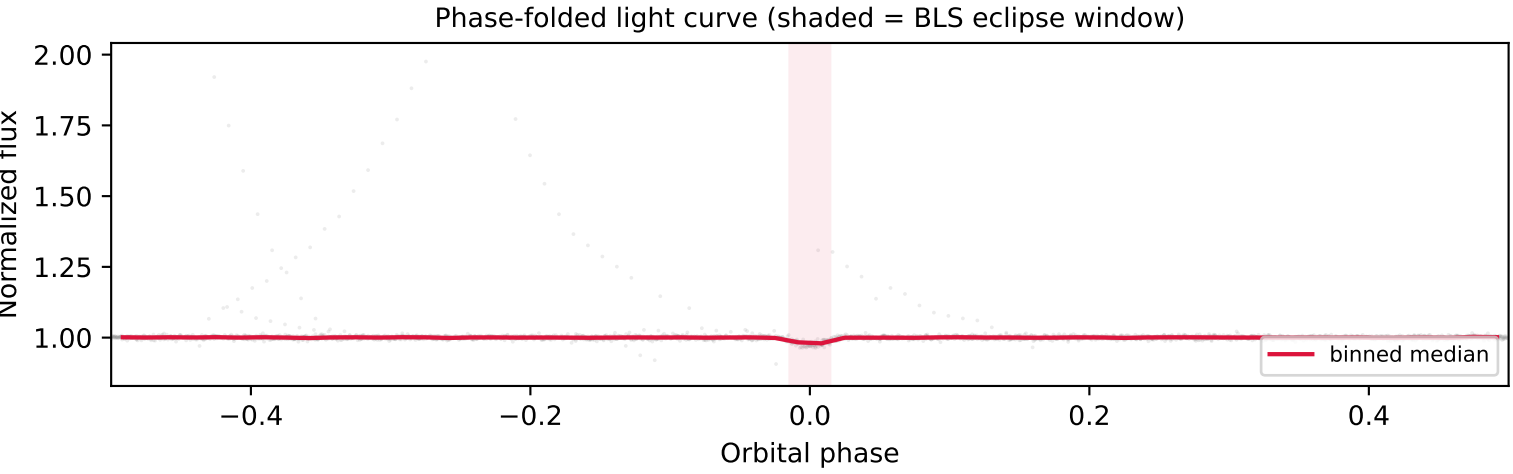
INDEPENDENT CHECKS (f3i gates)
  Eclipses observed     25  (gate >= 3)
  Robust depth ratio    0.91  (gate >= 0.50)
  Robust depth sigma    33.5  (gate >= 7.0)
  Odd/even z-score      0.46  (consistent)
  Secondary eclipse     not detected
```

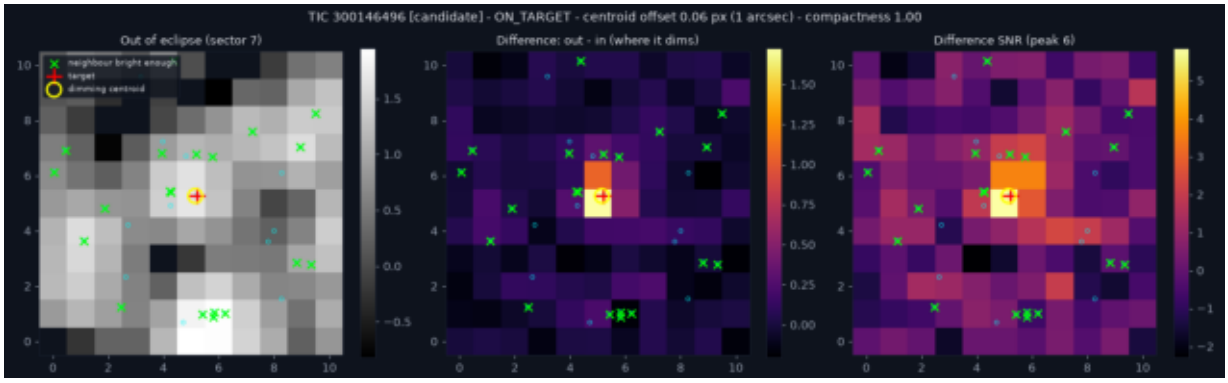
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CATALOG CROSS-MATCH (novelty)

  VSX                                no match
  ASAS-SN                            no match
  Gaia eclipsing-binary table        no match
  Gaia variable classification       no match
  TESS EB catalog (Prsa+2022)        no match
  ExoFOP TOI list                    no match

  Verdict: NOVEL
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```
PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)
  Sector 7    light-curve dip in this sector: 17.2 sigma    t0 refined by 0.0000 phase
  Difference-image peak SNR   5.84   (gate >= 5.0)      Compactness   1.00   (gate >= 0.50)
  Dimming centroid offset     0.06 px (1 arcsec)   (gate <= 1.0 px, i.e. on the target)
  Nearest bright-enough neighbour in cutout: TIC 300146493
    0.89 px away, dTmag +0.99 - farther from the dimming than the target
```





No caveats - clean detection on every check.